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1 #include <stdio.h>
2 #include <stdlib.h>
3 #include <pthread.h>
4 #include <unistd.h>
5 #include <semaphore.h>
6
7 #define NUM_PHILOSOPHERS 5
8
9 pthread_mutex_t chopsticks[NUM_PHILOSOPHERS];
10 sem_t room; // To avoid deadlock
11
12 void* philosopher(void* num) {
13     int id = *(int*)num;
14
15     while (1) {
16         // Thinking
17         printf("Philosopher %d is thinking\n", id);
18         sleep(rand() % 3 + 1);
19
20         // Enter room (max 4 philosophers)
21         sem_wait(&room);
22
23         // Pick chopsticks
24         pthread_mutex_lock(&chopsticks[id]);
25         pthread_mutex_lock(&chopsticks[(id + 1) % NUM_PHILOSOPHERS]);
26
27         // Eating
28         printf("Philosopher %d is eating\n", id);
29         sleep(rand() % 3 + 1);
30
31         // Put down chopsticks
32         pthread_mutex_unlock(&chopsticks[id]);
33         pthread_mutex_unlock(&chopsticks[(id + 1) % NUM_PHILOSOPHERS]);
34
35         // Leave room
36         sem_post(&room);
37     }
38 }
39
40 int main() {
41     pthread_t threads[NUM_PHILOSOPHERS];
42     int ids[NUM_PHILOSOPHERS];
43
44     // Initialize semaphore (only 4 allowed at a time)
45     sem_init(&room, 0, NUM_PHILOSOPHERS - 1);
46
47     // Initialize chopsticks
48     for (int i = 0; i < NUM_PHILOSOPHERS; i++) {
49         pthread_mutex_init(&chopsticks[i], NULL);
50     }
51
52     // Create threads
53     for (int i = 0; i < NUM_PHILOSOPHERS; i++) {
54         ids[i] = i;
55         pthread_create(&threads[i], NULL, philosopher, &ids[i]);
56     }
57
58     // Join threads
59     for (int i = 0; i < NUM_PHILOSOPHERS; i++) {
60         pthread_join(threads[i], NULL);
61     }
62
63     return 0;
64 }

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Philosopher 0 is thinking
Philosopher 4 is thinking
Philosopher 1 is thinking
Philosopher 2 is thinking
Philosopher 3 is thinking
Philosopher 1 is eating
Philosopher 1 is thinking
Philosopher 0 is eating
Philosopher 2 is eating
Philosopher 0 is thinking
Philosopher 4 is eating
Philosopher 2 is thinking
Philosopher 1 is eating
Philosopher 4 is thinking
Philosopher 3 is eating
Philosopher 1 is thinking
Philosopher 0 is eating
Philosopher 0 is thinking
Philosopher 3 is thinking
Philosopher 2 is eating
Philosopher 2 is thinking
Philosopher 1 is eating
Philosopher 1 is thinking
Philosopher 0 is eating
Philosopher 0 is thinking
Philosopher 4 is thinking
Philosopher 3 is eating
Philosopher 3 is thinking
Philosopher 2 is eating
Philosopher 2 is thinking
Philosopher 1 is eating
Philosopher 1 is thinking
Philosopher 0 is eating
Philosopher 0 is thinking
Philosopher 4 is eating
Philosopher 1 is eating
Philosopher 4 is thinking
Philosopher 3 is eating
Philosopher 1 is thinking
Philosopher 0 is eating
Philosopher 3 is thinking
Philosopher 2 is eating
Philosopher 0 is thinking
Philosopher 4 is eating
Philosopher 1 is eating
Philosopher 4 is thinking
Philosopher 3 is eating
Philosopher 2 is thinking
Philosopher 1 is eating
Philosopher 4 is thinking
Philosopher 3 is eating

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